FUNCTION GENERATOR

The function generator generates an output which is an arbitary function (with up to two points of inflection) of the input at JI1. This results in an effect that is similiar to but more complex and controllable than photographic solerization.

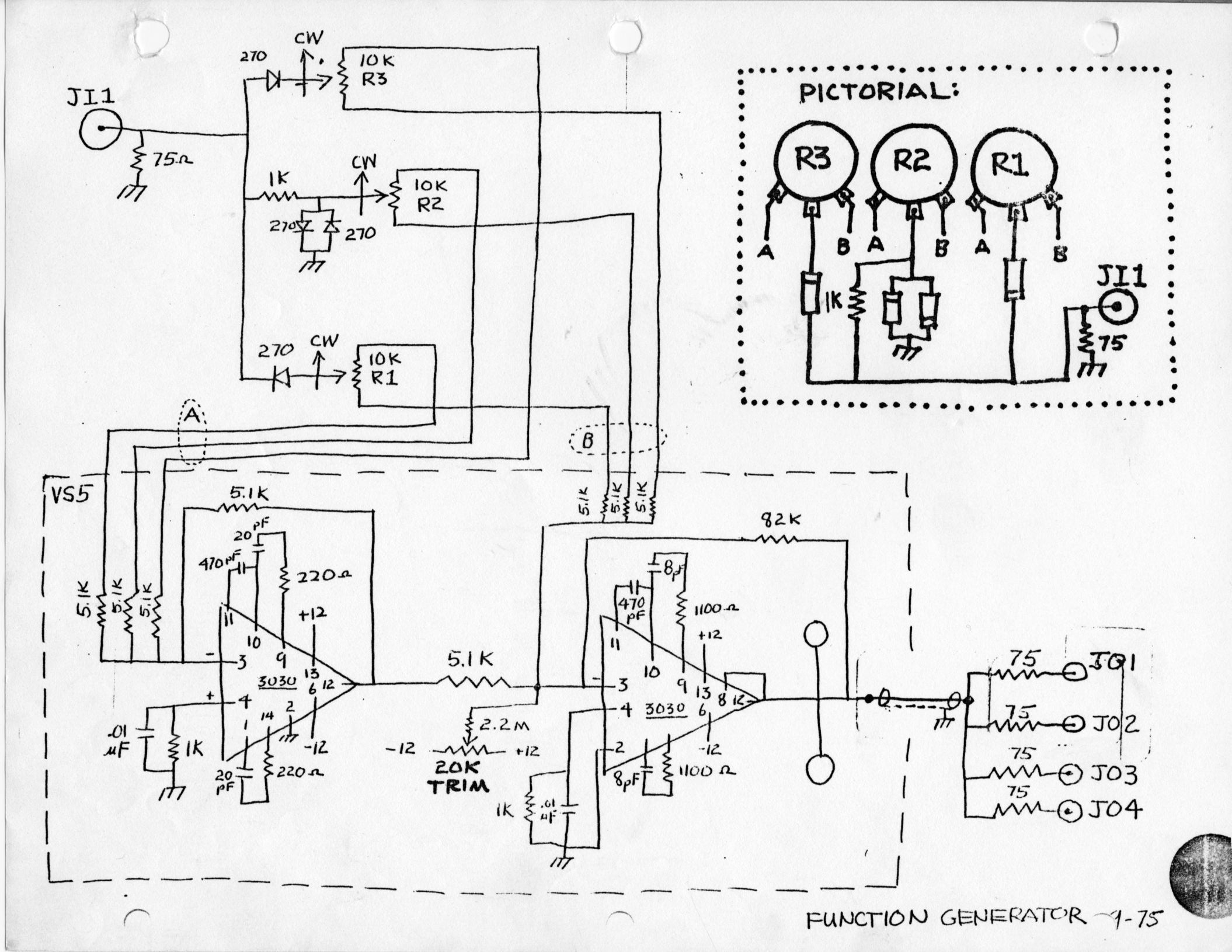
The function is controlled by R1, R2, and R3.
R1 controls the slope of the function for large negative inputs.
R2 controls the slope of the function for inputs near 0 voltages.
R3 controls the slope of the function for inputs of large positive voltage.

Clockwise is positive slope; counterclockwise is negative slope.

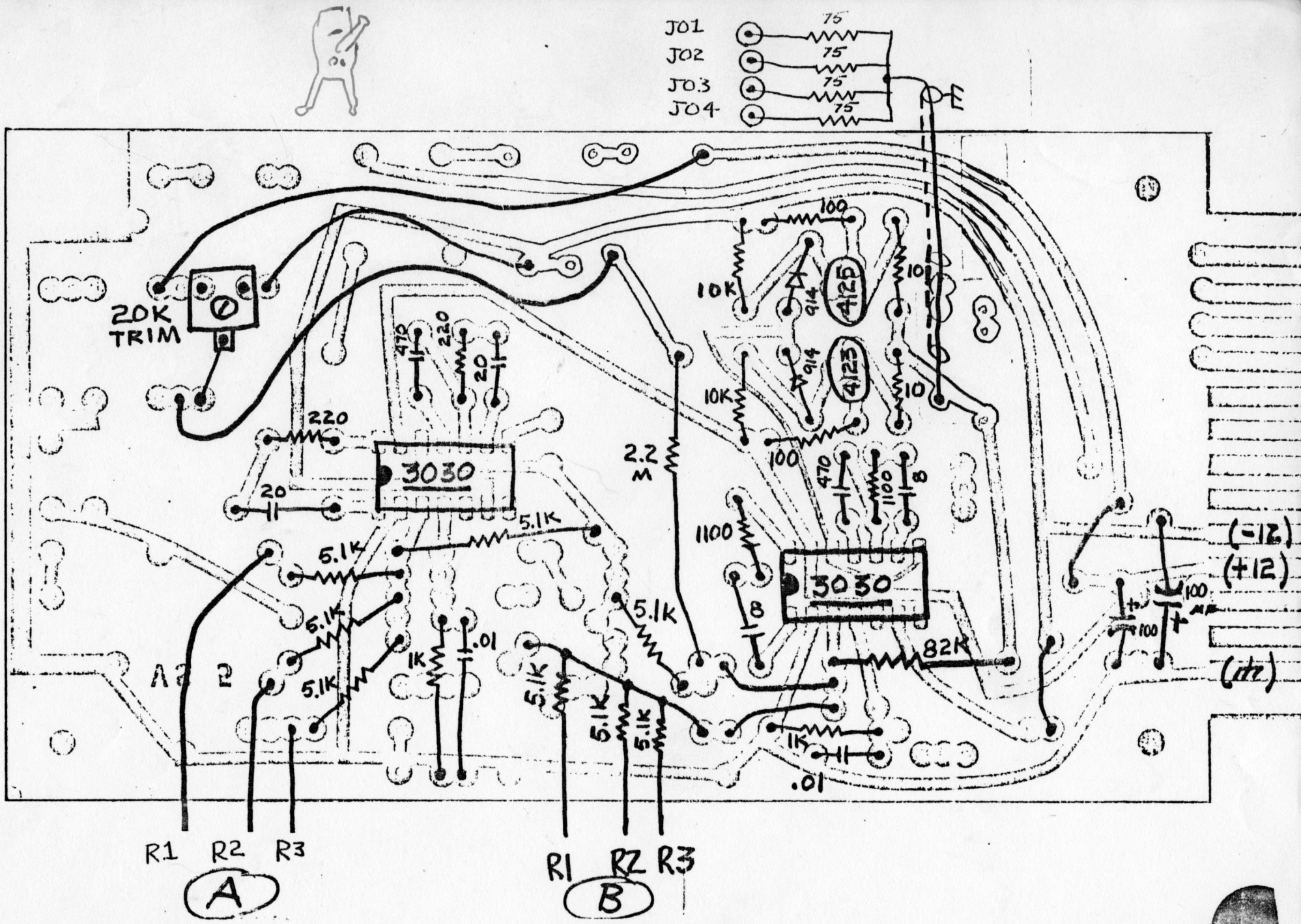
There are three electrical modules in one chassis box, so replicate work three times. Remember to buss (connect) +12 and -12 and ground wires from middle board to top and bottom board. Soldering directly to the foil is convenient.

TEST STUFF:

The 20K trimming resister on the VS5 board is adjusted such that no input results in 0 output voltage + or - .05 volts.




FUNCTION GENERATOR 1-75




FUNCTION GENERATOR 9-75

COMPARATOR

The comparator produces an output which is +.5 volts (white) if the input voltage at JI1 is greater (more positive) than the voltage at JI2.

The comparator produces an output which is -.5 volts (black) if the input voltage at JI1 is less (more negative) than the voltage at JI2.

With 0 volts or no input, the output will be either +.5 or -.5 volts into a 75 ohm load, depending on history.

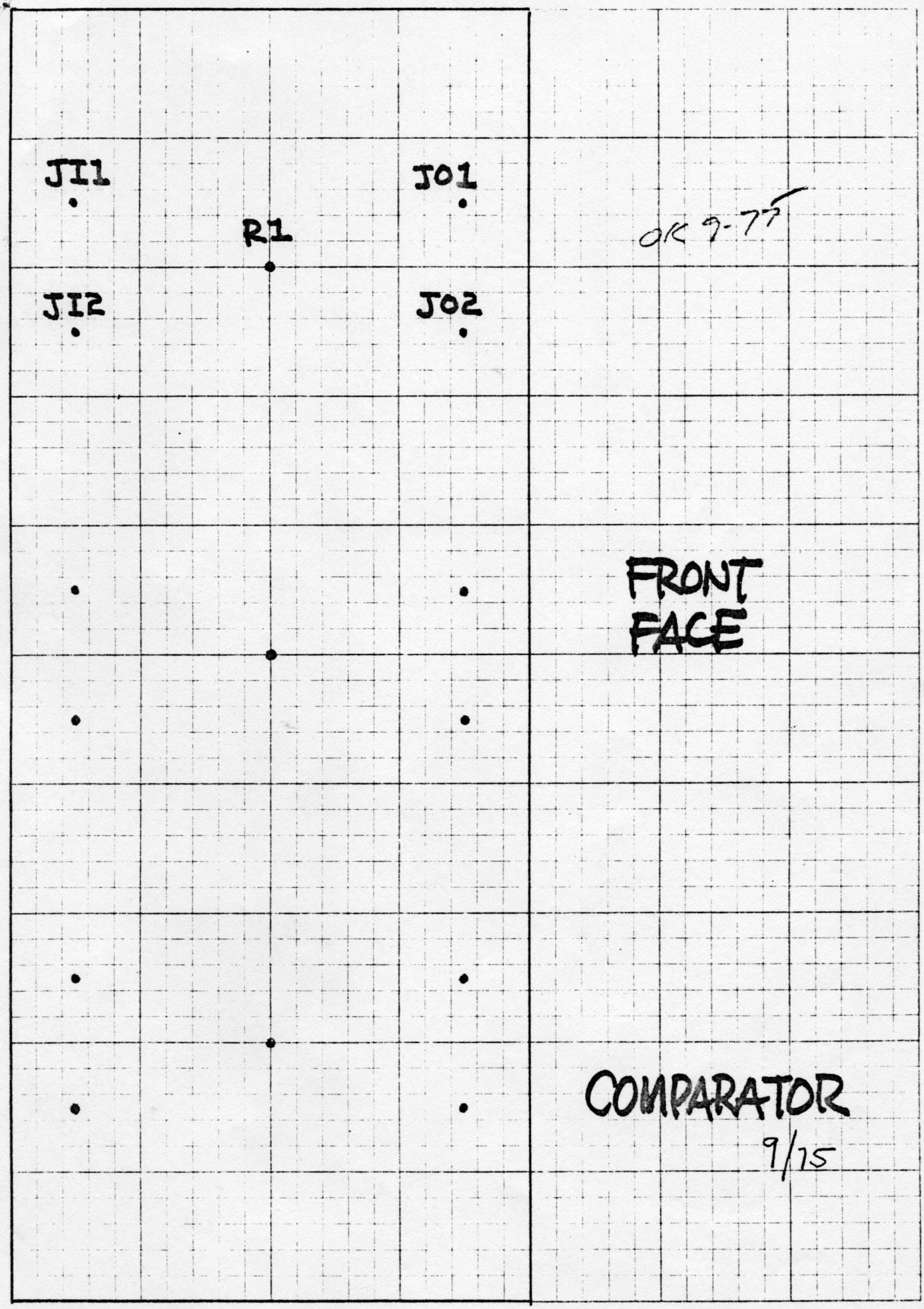
The variable resistor (pot) R1, determines the positive feedback which controls the tendency of the module to stay in the state it is in. Typically it is turned fully clockwise.

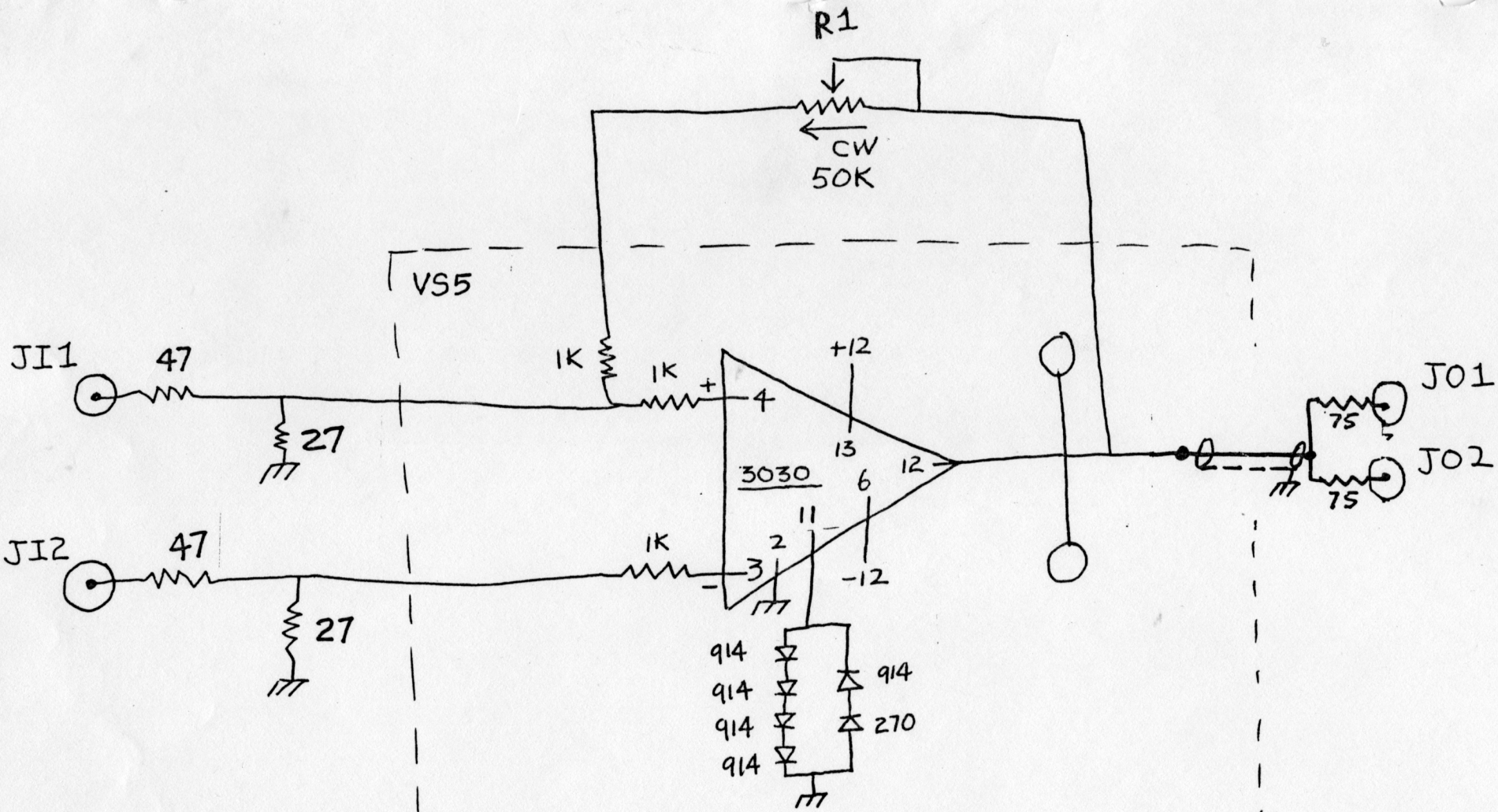
There are three electrical modules in one chassis box, so replicate work three times. Remember to buss (connect) + 12 volts, - 12 volts and ground from the center card to upper and lower cards.

TEST STUFF:

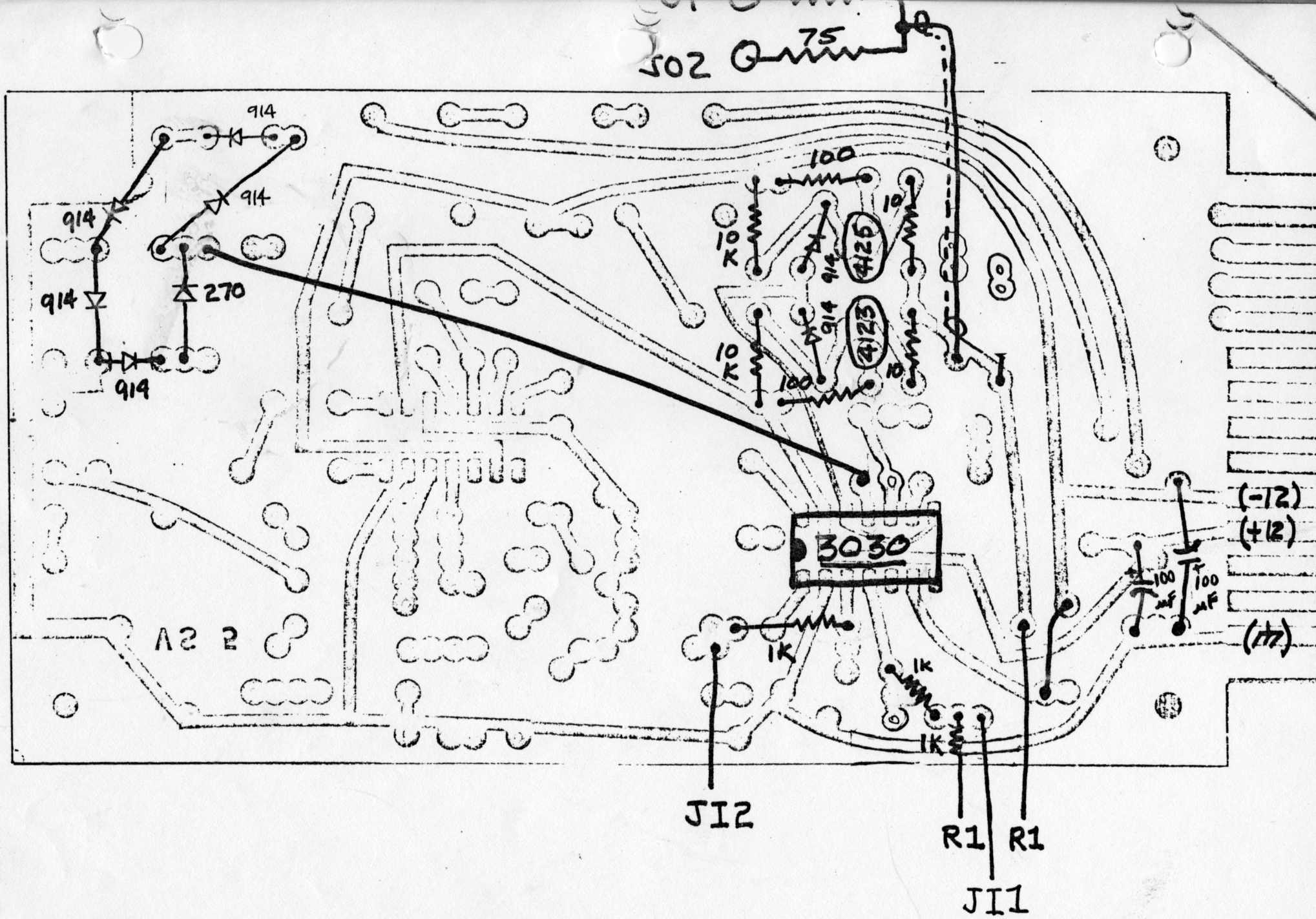
A sine-wave input should produce a clean square-wave output.

The output voltage should be between + or - .5 volts to + or - .75 volts.


JI1
J01
R1
OK 9-77
JI2
J02
FRONT
FACE
COMPARATOR
9/15




COMPARATOR 9-75




COMPARATOR 9-75

## AMPLITUDE CLASSIFIER

The amplitude classifier takes an input signal at JI 1land separates it into 8 contigeous regions varying from black to white. The value put out by each region is controled by R 1 through R 8 and by signals inputed to JI 1 through JI8. The output signals are available for each region seperately, JO 1 through JO 8. The sum of these signals is available at JO 21 through JO 24. The effect of JI 1, JI2, R 2, etc., is additive in each region. R11 controls the gain of the signal inputed at JI 11, and R 14 generates a bias (constant gray level proportional to knob position) which is added to the input signal. In general, R 11 and R 14 are used to match the incoming signal to the lightest, or 'top and 'darkest' or 'bottom' of the 8 regions.

## TEST INFORMATION

These tests are best performed with a 1 volt, peak to peak triangle wave inputed to JI 11, and a calibrated dual trace oscilloscope connected to the input and output of the amplitude classifier. R 13 is adjusted so that a +.4 volt signal activates channel 8 (bottom). R 12 is adjusted so that a +.4 volt signal activates channel 1 (top). R11 should be full clockwise and R 14 should be in the exact center of rotation. R 12 and R13 interact greatly, so, repeat adjustments until both conditions can be met simultaneously. R 16 should be adjusted so that with R 1 through R 8 in their centers the individual output are near 0 volts when not activated. R 15 should be adjusted so the summed output appearing at JO 21 through JO 24 is 0 volts for the non-activated channels.

## BASIC CIRCUIT DESCRIPTION AND TEST INFORMATION--AMPLITUDE CLASSIFIER

A signal at JI11 is sent to all comparators. The resistor string of 100 ohm resistors combined with voltage sources at R 13 and R 12 bias so that IC 1 switches on at a higher (+) voltage than IC 2, IC 2 switches on at a higher voltage than IC 3, etc.. For example; with 0 volts at JI 11, IC 5,6,7 might be on and IC 1,2,3,4 would be off. The transistors connected to the outputs of the comparators decode the comparator string output such that only the highest comparator on is outputed. In this example, channel 5 would be on, all others off. This signal is sent to the multiplier associated with channel 5, turning it on. All other multipliers would be off.